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・定積分の置換積分法 ～ $x = a\tan\theta$ と置換 ～

(例1) 次の定積分を求めよ。

(1) $\int_0^1 \frac{1}{x^2+1}dx$ (2) $\int_2^{2\sqrt{3}} \frac{1}{x^2+4}dx$ (3) $\int_0^{\sqrt{3}} \frac{1}{(x^2+3)\sqrt{x^2+3}}dx$

point
$\frac{1}{x^2+a^2}$ の定積分は、$x = a\tan\theta$ とおく

(1) $\int_0^1 \frac{1}{x^2+1}dx$

$x = \tan\theta$ とおくと

$dx = \frac{1}{\cos^2\theta}d\theta$

x	$0 \longrightarrow 1$
θ	$0 \longrightarrow \frac{\pi}{4}$

であるから

$$(与式) = \int_0^{\frac{\pi}{4}} \frac{1}{\tan^2\theta+1}\cdot\frac{1}{\cos^2\theta}d\theta$$
$$= \int_0^{\frac{\pi}{4}} \cos^2\theta\cdot\frac{1}{\cos^2\theta}d\theta \quad \leftarrow \ 1+\tan^2 x = \frac{1}{\cos^2 x}$$
$$= \int_0^{\frac{\pi}{4}} d\theta$$
$$= [\theta]_0^{\frac{\pi}{4}}$$
$$= \frac{\pi}{4}\ //$$

(2) $\int_2^{2\sqrt{3}} \frac{1}{x^2+4}dx$

$x = 2\tan\theta$ とおくと

$dx = \frac{2}{\cos^2\theta}d\theta$

x	$2 \longrightarrow 2\sqrt{3}$
θ	$\frac{\pi}{4} \longrightarrow \frac{\pi}{3}$

であるから

$$(与式) = \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{4(\tan^2\theta+1)}\cdot\frac{2}{\cos^2\theta}d\theta$$
$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos^2\theta}{4}\cdot\frac{2}{\cos^2\theta}d\theta$$
$$= \frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} d\theta$$
$$= \frac{1}{2}[\theta]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$
$$= \frac{\pi}{24}\ //$$

(3) $\int_0^{\sqrt{3}} \frac{1}{(x^2+3)\sqrt{x^2+3}}dx = \int_0^{\sqrt{3}} \frac{1}{(x^2+3)^{\frac{3}{2}}}dx$

$x = \sqrt{3}\tan\theta$ とおくと

$dx = \frac{\sqrt{3}}{\cos^2\theta}d\theta$

x	$0 \longrightarrow \sqrt{3}$
θ	$0 \longrightarrow \frac{\pi}{4}$

であるから

$$(与式) = \int_0^{\frac{\pi}{4}} \frac{1}{3^{\frac{3}{2}}(\tan^2\theta+1)^{\frac{3}{2}}}\cdot\frac{\sqrt{3}}{\cos^2\theta}d\theta$$
$$= \int_0^{\frac{\pi}{4}} \frac{\cos^3\theta}{3\sqrt{3}}\cdot\frac{\sqrt{3}}{\cos^2\theta}d\theta$$
$$= \frac{1}{3}\int_0^{\frac{\pi}{4}} \cos\theta\, d\theta$$
$$= \frac{1}{3}[\sin\theta]_0^{\frac{\pi}{4}}$$
$$= \frac{\sqrt{2}}{6}\ //$$

(例2) 定積分 $\int_0^1 \frac{1}{x^2-x+1}dx$ を求めよ。

$$(与式) = \int_0^1 \frac{1}{\left(x-\frac{1}{2}\right)^2+\frac{3}{4}}dx$$

$x-\frac{1}{2} = \frac{\sqrt{3}}{2}\tan\theta$ とおくと

$dx = \frac{\sqrt{3}}{2\cos^2\theta}d\theta$

x	$0 \longrightarrow 1$
θ	$-\frac{\pi}{6} \longrightarrow \frac{\pi}{6}$

であるから

$$(与式) = \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{1}{\frac{3}{4}(\tan^2\theta+1)}\cdot\frac{\sqrt{3}}{2\cos^2\theta}d\theta$$
$$= \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{4\cos^2\theta}{3}\cdot\frac{\sqrt{3}}{2\cos^2\theta}d\theta$$
$$= \frac{2\sqrt{3}}{3}\int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} d\theta$$
$$= \frac{2\sqrt{3}}{3}[\theta]_{-\frac{\pi}{6}}^{\frac{\pi}{6}}$$
$$= \frac{2\sqrt{3}}{9}\pi\ //$$